

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
8	(a)	(i)		award (1) for any of following • reactivity decreases down Group 7 • chlorine most reactive followed by bromine and then iodine • chlorine most reactive <u>and</u> iodine least reactive (accept reference to fluorine as most reactive) award (1) for any of following • it becomes harder to attract an electron • outer shell becomes further from the nucleus • more shielding	2			2		
		(ii)		award (1) for correct identification X chlorine Y iodine Z bromine award (1) for reasoning based on observations e.g. • chlorine reacts with sodium iodide and sodium bromide, (bromine reacts with sodium iodide only) and iodine reacts with none of the solutions • chlorine reacts with two solutions, (bromine reacts with one solution) and iodine reacts with none of them • two solutions change colour with chlorine, (one solution changes colour with bromine) and no solutions change colour with iodine • chlorine displaces bromine and iodine, (bromine displaces iodine) and iodine does not displace either of the other two			2	2		1

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
	(b)			65 (2) if answer incorrect award (1) for either of following 325 (mass of FeCl ₃ in equation) <i>M_r</i> FeCl ₃ = 162.5 alternative method award (1) for either of following 0.4 mol Fe reacts 0.4 mol FeCl ₃ forms ecf possible e.g. from incorrectly calculated <i>M_r</i> value		2		2	2	
	(c)			$\text{P}_4 + \boxed{6} \text{Cl}_2 \longrightarrow \boxed{4} \text{PCl}_3$		1		1		
				Question 8 total	2	3	2	7	2	1